

This is the definitive domain specification for the MH Catalog and Regulatory Reference Model. To maintain strict regulatory fidelity and future-proof the platform against the Ministry of Finance's (MH) evolving requirements, catalogs and schemas cannot be treated as static configurations. They must be modeled as first-class domain concepts. Here is the exhaustive, technology-agnostic design for how catalogs, schemas, and regulatory references exist within the domain.

1. Catalog Domain Model

Catalogs are modeled as an independent Bounded Context (or a dedicated module within the Type Registry) responsible for the exact definitions of all MH-mandated codes (e.g., Environment, Operation Types, Taxes, Payment Forms).

- **MHCatalog (Aggregate Root):** Represents a specific regulatory list.
 - **CatalogIdentifier:** The official MH designation (e.g., "Cat-002", "Cat-011").
 - **Name:** The human-readable name of the catalog.
 - **Description:** The regulatory purpose of the catalog.
- **CatalogEntry (Entity):** Represents a single valid option within a catalog.
 - **Code:** The exact string/alphanumeric value mandated by MH (e.g., "01", "13.00").
 - **Value:** The descriptive meaning (e.g., "Factura Electrónica", "Bienes").
 - **EffectivePeriod:** A Value Object containing ValidFrom and ValidTo dates.
 - **Status:** An enumerated state (Active, Deprecated, Future).
 - **Metadata:** A key-value collection for catalog-specific constraints (e.g., a tax catalog entry might have a metadata flag indicating if it applies to exports).

2. Catalog Versioning Strategy

MH catalogs evolve. Values are added, meanings change, and old codes are retired. The domain manages this exclusively through **Effective-Dating** rather than destructive updates.

- **Append-Only Immutability:** Existing CatalogEntry records are never overwritten or deleted.
- **Time-Bound Validity:** Every entry possesses an EffectivePeriod. When MH announces a new code, it is appended to the catalog with a future ValidFrom date.
- **Deprecation over Deletion:** When MH retires a code, the existing entry is not deleted. Its ValidTo date is populated, and its Status transitions to Deprecated.
- **Point-in-Time Resolution:** Any domain service querying a catalog must provide a ReferenceDate (typically the document issuance date). The model returns only the entries where the ReferenceDate falls within the EffectivePeriod.

3. Catalog Ownership Rules

Clear boundaries of ownership prevent cross-tenant contamination and ensure regulatory adherence.

- **System Ownership:** MHCatalogs are owned globally by the System/Platform. They are the single source of truth for all issuers.
- **Tenant Isolation:** Individual Issuers (Emisores) have **zero** authority to modify, add, or

delete MH catalog entries.

- **Issuer Mapping (Optional Overlay):** If an issuer needs to map their internal ERP codes (e.g., SKU categories) to an MH Catalog (e.g., Cat-011 "Tipo de Item"), this mapping is owned by the Catalog Context of the Issuer. The mapping references the CatalogEntry identity, but the MH Catalog itself remains untouched.

4. Catalog Update Lifecycle

When a regulatory mandate introduces catalog changes, the domain handles the update through a strict state machine.

State	Trigger & Description
Draft	A new entry is staged in the system but is invisible to validation engines. Used for preparation before the legal effective date.
Active	The entry's ValidFrom date is reached. It is now legally valid for use in new documents.
Deprecated	The entry's ValidTo date has passed. It can no longer be selected for new documents but remains in the system to validate historical documents and ensure audit integrity.

5. Regulatory Reference Model

How the rest of the domain interacts with the catalogs.

- **By-Value Copying (Snapshots):** When a FiscalDocument is composed and validated, it does not maintain a live pointer to a CatalogEntry. It copies the exact Code and Value into its own immutable Payload/Snapshot.
- **Severed Links:** Once a document transitions to Signed, its relationship to the active catalog is conceptually severed. If MH updates the catalog tomorrow, the signed document remains perfectly intact and legally accurate for its time of issuance.
- **Cross-Context Validation:** The Issuance context requests a validity check from the Catalog context by passing the selected code, the catalog identifier, and the document's issuance timestamp.

6. Schema Version Reference Model

MH periodically releases entirely new JSON schema structures (e.g., moving from v1 to v3). The domain treats the Schema Version as a first-class governing object.

- **SchemaVersionBinding (Aggregate Root):** Defines the exact structural and regulatory

requirements for a specific DTE Type at a specific point in time.

- **DTEType:** The document type (e.g., 01, 03).
- **VersionString:** The official MH version label (e.g., "v1", "v3").
- **EffectivePeriod:** The dates during which MH accepts this schema.
- **MandatedCatalogs:** A defined list of CatalogIdentifiers that must be enforced for this schema version.

7. Validation Reference Model

Validation rules are not hard-coded into the FiscalDocument. They are projected through the SchemaVersionBinding.

- **Rule Encapsulation:** Validation logic is encapsulated in discrete ValidationRule Value Objects attached to the SchemaVersionBinding.
- **Context-Aware Execution:** When a document requests validation, the engine resolves the active SchemaVersionBinding for that DTEType and executes the attached rules.
- **Catalog-Driven Rules:** Rules explicitly reference catalogs. A rule might state: *"The attribute Environment must exist within Cat-002 where status is Active relative to the current timestamp."*

8. Backward Compatibility Rules

The platform must maintain the ability to read, render, and prove the legal validity of a document years after the schemas and catalogs have changed.

- **Permanent Resolution:** The system must always be able to resolve what a catalog entry meant on a specific date in the past. The append-only catalog model guarantees this.
- **Schema Pinning:** Every FiscalDocument permanently records the VersionString it was validated and signed against.
- **Evidence Independence:** The EvidenceBundle containing the raw signed payload and the MH response is the ultimate truth. It relies on cryptographic hashes, not on the current state of the regulatory catalogs.

9. Multi-Version Support Strategy

MH frequently provides a "transition window" where both old and new schema versions are accepted simultaneously.

- **Concurrent Bindings:** The domain supports multiple SchemaVersionBindings for the same DTEType being Active simultaneously, provided their EffectivePeriods overlap.
- **Default Targeting:** The IssuerConfiguration aggregate maintains a TargetSchemaVersion setting. This determines which schema version is used by default when composing a new document.
- **Gradual Rollout:** This allows the platform owner to transition specific issuers to a new MH schema (e.g., v3) while leaving others on the older schema (e.g., v1) until the legal deadline, simply by updating the issuer's configuration setting.